| INCH-POUND |

ZZ-H-521F
May 28, 1993
SUPERSEDING
ZZ-H-521E
October 1, 1987

FEDERAL SPECIFICATION

HOSE AND HOSE ASSEMBLIES, NONMETALLIC, SPRAY

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers textile reinforced, nonmetallic spray hose and hose assemblies. These items are to be used with spray equipment and will convey oil emulsions, paint, dope, and solvents.

1.2 Classification. The hose shall be of the following sizes as specified (see 6.2):

Inside diameter (inch) - 0.25, 0.31, 0.38, 0.50, 0.75, and 1.00

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 621 Pleasant Valley Road, Port Hueneme, CA 93043-4300, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4720

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Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for
FED-STD-601 - Rubber: Sampling and Testing

Military Specification

MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware: Packaging of

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-889 - Dissimilar Metals

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

ASTM

ASTM D380 - Standard Test Methods for Rubber Hose
ASTM D412 - Standard Test Methods for Rubber Properties in Tension
ASTM D413 - Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate
ASTM D471 - Standard Test Method for Rubber Property - Effect of Liquids
ASTM D573 - Standard Test Method for Rubber Deterioration in an Air Oven
ASTM D638 - Standard Test Method for Tensile Properties of Plastics
ASTM D882 - Standard Test Methods for Tensile Properties of Thin Plastic Sheeting
ASTM D1149 - Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.2 Construction. The hose shall conform to the physical properties and pressure requirements specified in table I without evidencing leakage, cracking, rupture, separation of hose fittings, or other defects.

TABLE I. Physical requirements.

Size (inside diameter) (inch)	0.25 0.31	0.38	0.50	0.75	1.00
Tolerance inside diameter, inch, plus or minus	0.03	0.03	0.03	0.03	0.06
Thickness, inch					
Cover, minimum	0.023	0.023	0.023	0.023	0.023
Tensile strength (before aging)					
Tube, minimum, psi	800	800	800	800	800
Cover, minimum, psi	1250	1250	1250	1250	1250
Decrease tensile strength (after aging)					
Tube maximum, percent	35	35	35	35	35
Cover maximum, percent	35	35	35	35	35
Elongation, ultimate (before aging)					
Tube minimum, percent	200	200	200	200	200
Cover minimum, percent	250	250	250	250	250
Elongation, ultimate (after aging) decrease					
Tube maximum, percent	55	55	55	55	55
Cover maximum, percent	55	55	55	55	55
Adhesion - pounds/inch width					
Tube to plies minimum	6	6	6	6	6
Between plies minimum	10	10	10	10	10
Cover to plies minimum	6	6	6	6	6
Reinforcement, plies minimum					
Wrapped	2	2	2	2	2
Braided	1	1	1	1	1
Helically wound (spiral)	2	2	2	2	2
Hydrostatic test pressure, minimum, psi					
Coupling test*	400	400	400	400	400
Proof test**	400	400	400	400	400
Burst test	800	800	800	800	800

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*This test shall be made only when hose is purchased with couplings. The specified pressure shall be maintained for 2 minutes.

**When ordered without couplings, the hose shall be proof tested by the manufacturer in production lengths at 400 pound-force per square inch (psi) for 1 minute.

3.2.1 Wrapped hose. Wrapped hose shall consist of a smooth bore tube, plies of cotton or synthetic fiber fabric, and a smooth cover.

3.2.2 Braided hose. Braided hose shall consist of a smooth bore tube, one or more braids of cotton or suitable synthetic fiber yarn or cord, and a smooth cover.

3.2.3 Helically wound (spiral) hose. Helically wound hose shall consist of a smooth bore tube, two or multiples of two helically wound plies (a ply shall be wound in a direction opposite to that of the adjacent ply) of cotton or synthetic fiber yarn or cord, and a smooth cover.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance of maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the product covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.3.1 Tube. The tube shall consist of either a suitable thermoplastic material or a suitable synthetic rubber compound and shall meet the requirements of table I as well as the tests for solvent resistance (see 4.5.7).

3.3.2 Cover. The cover shall consist of a suitable synthetic rubber compound and shall meet the requirements of table I as well as the tests for solvent and ozone resistance (see 4.5.7 and 4.5.8).

3.3.3 Material deterioration prevention and control. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the hose assemblies may be exposed.

3.3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish of treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer of designated representative.

3.4 Reinforcement. The fabric or yarn reinforcement shall be of sufficient strength to meet the hydrostatic burst test requirements of table I.

3.5 Couplings. When required (see 6.2), couplings and threads shall be of the type specified. The threads shall conform to FED-STD-H28.

3.5.1 Permissible bulge. When hose couplings are provided, the maximum permissible bulge of the inner tube shall be as shown in figure 1 and as indicated by the minimum diameter "B" below:

Inside diameter (inches)	"B" (inches)	Inside diameter (inches)	"B" (inches)
0.25	0.20	0.50	0.45
0.31	0.26	0.75	0.70
0.38	0.33	1.00	0.90

3.6 Marking. Each length of hose shall be marked in one of the following manners:

a. By inlaying a rubber or other suitable material brand and by vulcanizing said brand into the cover so that it cannot be removed except by mechanical means. The brands shall be a color that contrasts with the color of the cover and shall be located at approximate 12.5 feet intervals on braided hose and at two places on each length, approximately 4 feet from the ends, on wrapped hose.

b. By printing with suitable composition ink in a continuous line along the hose and vulcanizing said ink into the cover. The ink shall be of a color that contrasts with the color of the cover. The wording specified below shall be repeated at maximum intervals of 36 inches.

c. By applying a continuous embossed strip (plastic or metallic) along the hose and embedding the strip into the outer surface of the cover during vulcanization so as to produce the wording specified below in relief along the entire length. No color contrast is necessary in this means of marking. The required marking shall be repeated at maximum intervals of 36 inches.

The legend required for any means of marking shall consist of the manufacturer's name and/or recognized trademark; the word "SPRAY" plus the quarter and year of manufacture (e.g., 2Q77). The minimum letter height shall be 0.12 inch for any means of marking.

3.7 Lengths. The hose shall be furnished in the lengths specified (see 6.2), with 50 feet being the maximum. Couplings shall be considered additions to the hose length, not part of the length of the hose. A tolerance of ±1 percent shall be permitted in the length of the hose.

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3.8 Tensile strength and ultimate elongation after immersion.

3.8.1 Tube. After subjection to the immersion test described in 4.5.6, the tensile strength of the tube shall be at least 500 psi, and the decrease from the original elongation shall not exceed 40 percent.

3.8.2 Cover. After subjection to the immersion test described in 4.5.6, the tensile strength of the cover shall be at least 650 psi, and the decrease from the original tensile strength shall not exceed 50 percent. The decrease from the original elongation shall not exceed 45 percent.

3.9 Resistance to solvents.

3.9.1 Change in volume.

3.9.1.1 Tube. The increase in volume of the tube after its immersion in each of the following solvents by the procedure described in 4.5.7.1 shall not exceed the value given for the respective solvent. In case the specimen dissolves or shrinks in volume more than 5 percent, the sample shall be considered to have failed to meet the requirements.

Solvent	Increase in volume (percent maximum)
Ethyl acetate (CP)	30
Acetone (CP)	30
Water, distilled	6

3.9.1.2 Cover. The increase in volume of the cover when immersed in the testing fuel specified in 4.5.6 and tested as described in 4.5.7.2 shall not exceed 100 percent.

3.9.2 Extraction. The extractable material in the tube compound, after its immersion in each of the following solvents by the procedure described in 4.5.7.3, shall not exceed the value shown for the respective solvent.

Solvent	Extractable material by Weight (percent maximum)
Ethyl acetate (CP)	8
Acetone (CP)	20
Water, distilled	6

3.10 Accelerated aging. The decrease in tensile strength and ultimate elongation of the tube and cover compounds, after exposure to the test described in 4.5.4, shall not exceed the values given in table I.

3.11 Ozone resistance. When subjected to the test specified in 4.5.8, the cover of the hose shall show no breaks or cracks.

3.12 Blockage. When tested as specified in 4.5.1, the hose shall show no blocking, webbing, or bridging across its inner diameter.

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3.13 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce the type of hose specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. First article inspection shall be performed on one sample hose length when a first article sample is required (see 3.1, 6.2, and 6.6). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be a standard production item from the contractor's current inventory provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one hose length or hose assembly. All hose and hose assemblies offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is

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rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots. Guidance for sampling and acceptance procedures is provided in 6.5.

4.3.1 Sampling for tests. A representative sample 3 feet in length shall be cut from lengths selected from each lot. If hose is ordered with couplings (see 3.5), the supplier shall reattach couplings to the remaining length from which the 3-foot length was cut. After the couplings have been reattached, the remaining length of the hose shall be used to conduct the hydrostatic coupling test specified in 4.5.5.2. When the remaining length has passed the hydrostatic coupling test, it shall be accepted as a full length of hose.

4.4 Examination. Each of the sample hose lengths selected shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection.

4.5 Tests. Sample hose sections selected shall be subjected to the applicable test specified in table II and 4.5.1 through 4.5.8.

4.5.1 Blockage test. The sample hose shall be tested for blocking, webbing, or bridging across its inner diameter. A steel ball, 0.125 inches smaller in diameter than the nominal inside diameter of the hose shall be used for testing. The ball shall be rolled or blown by low pressure through the hose. Any hose that cannot pass the ball freely shall be considered a defective unit.

4.5.2 Tensile strength and ultimate elongation. The tensile strength and elongation of the tube and cover shall be determined as described in ASTM D412, D638, or D882 as appropriate. Die B shall be used for preparing ASTM D412 test specimens. The speed of testing shall be 2 inches per minute for ASTM D638. Plate specimens may be used for testing tensile strength and elongation of tube in 0.25, 0.31, and 0.38 inch inside diameter sizes when thickness of tube is below 0.039 inch to reduce variation in test values. Plate specimens shall be taken from the same material as the tube. Plate specimens shall not be more than 0.125 inch thick.

TABLE II. Test method references.

Test	Test Method No.	Reference paragraph
Length	ASTM D380	3.7
Hydrostatic burst	ASTM D380	4.5.5.1
Hydrostatic coupling test	ASTM D380	4.5.5.2
Adhesion	ASTM D413	4.5.3
Extraction solvent	6511*	4.5.7.3
Volume swell solvent	ASTM D471	4.5.7.1

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TABLE II. Test method references. - continued

Tensile strength and ultimate elongation (synthetic rubber compound)	ASTM D412	4.5.2
Tensile strength and ultimate elongation (thermoplastic material)	ASTM D638, D882	4.5.2
Tensile strength and ultimate elongation after immersion	ASTM D471	4.5.6
Aging accelerated	ASTM D573	4.5.4
Ozone resistance	ASTM D1149	4.5.8

*Method in FED-STD-601.

4.5.3 Adhesion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other shall be determined as described in ASTM D413. The specimen shall be the ring type.

4.5.4 Accelerated aging. The resistance of the tube and cover to accelerated aging shall be determined in accordance with ASTM D573, except that the time of aging shall be 70 hours at a temperature of 212 degrees Fahrenheit (^{o}F). Tensile strength and ultimate elongation tests shall be performed to determine the requirements of table I.

4.5.5 Hydrostatic tests.

4.5.5.1 Burst. The hydrostatic burst test shall be determined as described in ASTM D380. The specimen shall be 3 feet in length. The hose assembly from which a 3-foot sample has been taken for burst test shall have the couplings refitted by the supplier and the assembly shall be considered a full length, if the hose is found to be in compliance with this specification.

4.5.5.2 Coupling. The hose assembly shall be tested as described in ASTM D380, except that the pressure shall be as indicated in table I, and shall be maintained for at least 2 minutes.

4.5.6 Tensile and elongation after immersion. The tensile strength and ultimate elongation after immersion shall be tested as described in ASTM D471. The test fluid shall conform to Reference Fuel D as specified in ASTM D471.

4.5.7 Resistance to solvents.

4.5.7.1 Swelling test - tube. The change in volume (swelling) of the tube shall be tested as described in ASTM D471. The test fluids shall be as specified in 3.9.1.1.

4.5.7.2 Swelling test - cover. The change in volume (swelling) of the cover shall be tested as described in ASTM D471. The test fluids shall be as specified in 3.9.1.2 and 4.5.6.

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4.5.7.3 Extraction. The extractable material of the tube shall be tested as described in method 6511 of FED-STD-601, except that the extract shall be dried at a temperature of 122°F ±3.6°F. The solvents shall be as specified in 3.9.2.

4.5.8 Ozone resistance. The sample hose cover shall be tested for ozone resistance in accordance with ASTM D1149, except that after conditioning for 24 hours in an ozone free atmosphere, the stretched sample shall be exposed for 72 hours at 104°F ±3.6°F to an atmosphere containing 50 parts per hundred million of ozone.

4.6 Preparation for delivery inspection. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one unit prepared for shipment, i.e., one shipping container, one reel, or one hose length as applicable. Sampling shall be in accordance with MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation and packaging shall be in accordance with the Level A or Level C requirements of MIL-H-775, as specified (see 6.2). Packing shall be in accordance with the Level A, B, or C requirements of MIL-H-775, as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The spray hose is intended for use in base and field shops for spraying purposes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in acquisition documents:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. First article sample, when required for inspection and approval (see 3.1, 4.2.1, and 6.6).
e. Type of couplings and threads (see 3.5).
f. Length required (see 3.7).
g. Level of preservation/packaging and level of packing required (see 5.1).

6.3 Recommended working pressure. Recommended working pressure for the hose covered by this specification is 200 psi.

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6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 Sampling procedures.

6.5.1 Sampling procedures for examination. Recommended inspection level is II and Acceptable Quality Level (AQL) is 2.5 percent defective for major defects and 4.0 percent defective for minor defects (see 4.3).

6.5.2 Sampling procedures for tests. Recommended inspection level is S-3 and AQL is 4.0 percent defective (see 4.3).

6.6 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete hose length or hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.7 Part or Identifying Number (PIN). The PIN to be used for hose and hose assemblies acquired to this specification is derived by selecting from the requirement options available in this specification as follows:

Hose assembly:

Specification part number	ZZ-H-521	XXX	XXX
Specification number			
Hose size code number (see 6.7.1)			
Length in inches (see 3.7)			

6.7.1 Hose size. Hose size is designated by a code number (see table III).

TABLE III. Hose size code number.

Hose size code number	025	031	038	050	075	100
Hose ID (inches)	0.25	0.31	0.38	0.50	0.75	1.00

6.8 Subject term (key word) listing.

Paint
Solvents
Spraying

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MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

Air Force - 82
DLA - CS

User Activities

Navy - CG, MC, SA

CIVIL AGENCY COORDINATING ACTIVITIES:

HHS - FEC
GSA - FSS

PREPARING ACTIVITY:

Navy - YD

(Project 4720-0029)

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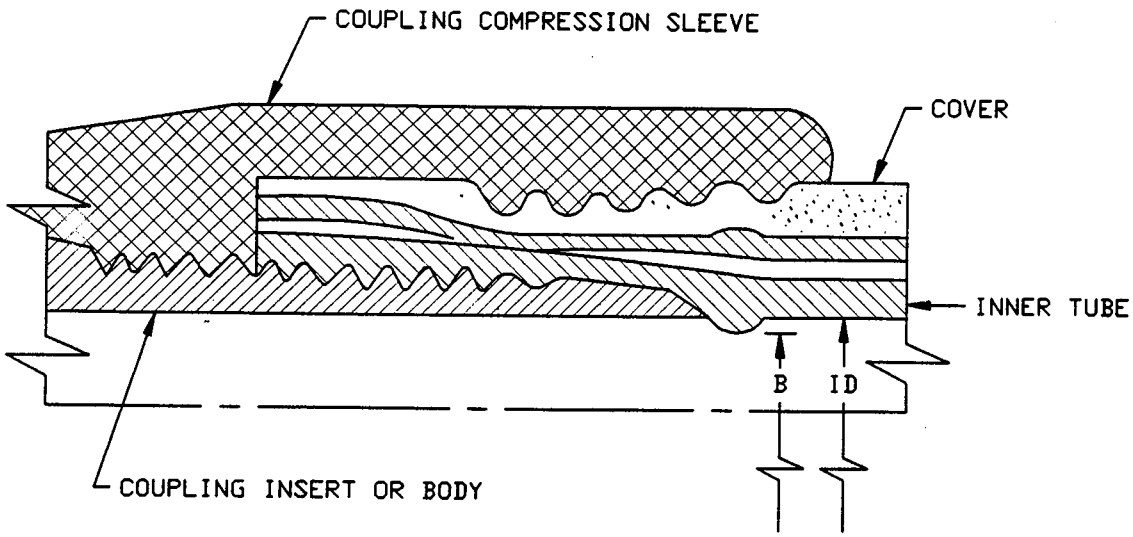


FIGURE 1. Hose dimensions.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER ZZ-H-521F	2. DOCUMENT DATE (YYMMDD) 930507

3. DOCUMENT TITLE
HOSE AND HOSE ASSEMBLIES, NONMETALLIC, SPRAY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
DR. H. A. PORTE	805-982-5782	551-5782
c. ADDRESS (Include Zip Code) COMMANDING OFFICER, NCBC CODE 1564H 621 PLEASANT VALLEY ROAD PORT HUENEME, CA 93043-4300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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